

Two Sum + Array is Sorted

→ Bruteforce

```
for (i = 0 → n) {  
  for (j = 0 → n) {  
    if (arr[i] + arr[j] == Target) return {i, j};  
  }  
}
```

T.C → $O(n^2)$

T.C is too high

→ Optimal

T.C → $O(n)$
S.C → $O(1)$

(Two pointer Approach)

```
s = 0, e = n - 1  
while (s < e) {  
  sum = arr[s] + arr[e];  
  if (sum == Target) return {s, e};  
  else if (sum < Target) s++;  
  else e--;  
}
```

return {-1, -1}

Dry Run :-

arr → [2, 3, 7, 9]
s → 0, e → 3

sum = 2 + 9 = 11 > 9
sum = 3 + 9 = 12 > 9
sum = 7 + 9 = 16 > 9
sum = 2 + 7 = 9 == 9 return {s, e};

Target = 9